

WORKOPS - SMART COMPANY MANAGEMENT SYSTEM

COMPLETE DATABASE DOCUMENTATION

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SECTION 1: DATABASE OVERVIEW

1.1 Database Identity

Attribute	Value
Database Name	WorkOpsDB
Server Instance	HPVICTUS\MSSQLSERVER02
Authentication Mode	Windows Authentication
Created For	WorkOps Smart Company Management System
Database Version	SQL Server 2012 or higher
Character Set	Latin1_General_CI_AS

1.2 Database Statistics

Metric	Count
Total Tables	14
Total Stored Procedures	5
Total Views	4
Total Triggers	4

Total Functions	3
Total Indexes	25+
Total Foreign Keys	20+

1.3 Database Purpose

The WorkOps database serves as the central data repository for the entire company management system. It stores all information related to:

- User authentication and role management
- Employee personal and professional information
- Organizational department structure
- Task assignment and tracking
- Daily attendance records
- Leave applications and balances
- System notifications and activity logs
- Company holidays and system settings
- Performance review records

SECTION 2: DATABASE NAMING CONVENTIONS

2.1 Table Naming Convention

Rule	Description	Example
Format	PascalCase, plural nouns	Employees, Departments
Prefix	No prefix (not tbl_)	WorkTasks (not tbl_WorkTasks)

Junction Tables	Combine two table names	TaskDocuments, LeaveBalances
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2.2 Column Naming Convention

Rule	Description	Example
Format	PascalCase	FirstName, EmployeeID
Primary Key	TableName + ID	EmployeeID, DepartmentID
Foreign Key	RelatedTableName + ID	DepartmentID, RoleID
Boolean Columns	Is prefix	IsActive, IsPaid, IsRead
Date Columns	Date suffix for dates	JoiningDate, CreatedDate
DateTime Columns	No special suffix	CheckInTime, AppliedDate

2.3 Constraint Naming Convention

Constraint Type	Prefix	Example
Primary Key	PK_	PK_Employees
Foreign Key	FK_	FK_Employees_Departments
Unique Constraint	UQ_	UQ_Users_Email
Check Constraint	CHK_	CHK_WorkTasks_Priority

Default Constraint	DF_	DF_Employees_CreatedDate
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2.4 Index Naming Convention

Index Type	Prefix	Example
Non-clustered	IX_	IX_Employees_Email
Unique Non-clustered	UX_	UX_Users_Email

2.5 Stored Procedure Naming Convention

Rule	Description	Example
Format	sp_ prefix + PascalCase	sp_AuthenticateUser
Purpose	Describes action	sp_GetAdminDashboardData

2.6 View Naming Convention

Rule	Description	Example
Format	vw_ prefix + PascalCase	vw_EmployeeDetails
Purpose	Describes data	vw_MonthlyAttendance

2.7 Function Naming Convention

Rule	Description	Example
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Format	fn_ prefix + PascalCase	fn_CalculateExperience
Purpose	Describes calculation	fn_GetPriorityColor

2.8 Trigger Naming Convention

Rule	Description	Example
Format	trg_ prefix + TableName + Action	trg_Employees_CreateLeaveBalance
Action	AFTER INSERT, UPDATE, DELETE	trg_WorkTasks_ActivityLog

SECTION 3: COMPLETE DATA DICTIONARY

3.1 Data Types Used

Data Type	Usage	Example Columns
INT	Primary keys, foreign keys, counters	EmployeeID, RoleID, Status
BIGINT	Large logs, audit tables	LogID
NVARCHAR(50)	Short text, codes, names	FirstName, DepartmentCode
NVARCHAR(100)	Email, medium text	Email, Position

NVARCHAR(200)	Titles, names	Title, DepartmentName
NVARCHAR(255)	Hashed passwords, file paths	PasswordHash, ProfilePicture
NVARCHAR(500)	Addresses, descriptions	Address, Description
NVARCHAR(MAX)	Long text, comments	CommentText, Reason
DATETIME	Timestamps with time	CheckInTime, CreatedDate
DATE	Dates only	JoiningDate, AttendanceDate
DECIMAL(18,2)	Monetary values	Salary
DECIMAL(5,2)	Hours, percentages	TotalHours, OvertimeHours
BIT	Boolean flags	IsActive, IsPaid, IsRead

SECTION 4: TABLE-BY-TABLE DETAILED DOCUMENTATION

4.1 Roles Table

Table Description

The Roles table stores the different user roles available in the system. Each role defines a set of permissions and access levels. The system uses three default roles: Admin, Manager, and Employee.

Table Schema

Column Name	Data Type	Nullable	Default	Description
RoleID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each role
RoleName	NVARCHAR(50)	No	None	Name of the role (Admin, Manager, Employee)
Description	NVARCHAR(200)	Yes	NULL	Detailed description of role permissions
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when the role was created
IsActive	BIT	Yes	1	Indicates if the role is active and assignable

Constraints

Constraint Name	Type	Columns	Description
PK_Roles	Primary Key	RoleID	Unique identifier for each role
UQ_Roles_RoleName	Unique	RoleName	Role names must be unique
DF_Roles_CreatedDate	Default	CreatedDate	Default to current date/time

DF_Roles_IsActive	Default	IsActive	Default to active (1)
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Sample Data

RoleID	RoleName	Description	IsActive
1	Admin	System Administrator with full access	1
2	Manager	Department Manager with team management access	1
3	Employee	Regular employee with limited access	1

Relationships

Foreign Key	Referenced Table	Referenced Column
FK_Users_Roles	Users	RoleID

4.2 Users Table

Table Description

The Users table stores authentication credentials for all system users. This table is separate from Employees to maintain security and allow for users who may not be employees (future expansion). Passwords are stored as SHA256 hashes with salt.

Table Schema

Column Name	Data Type	Nullable	Default	Description
UserID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each user
Email	NVARCHAR(100)	No	None	User's login email address (unique)
PasswordHash	NVARCHAR(255)	No	None	SHA256 hashed password with salt
RoleID	INT	No	None	Foreign key to Roles table
IsActive	BIT	Yes	1	Indicates if user account is active
LastLoginDate	DATETIME	Yes	NULL	Timestamp of most recent successful login
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when user account was created
FailedLoginAttempts	INT	Yes	0	Counter for consecutive failed login attempts

IsLocked	BIT	Yes	0	Indicates if account is locked due to failed attempts
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Constraints

Constraint Name	Type	Columns	Description
PK_Users	Primary Key	UserID	Unique identifier for each user
UQ_Users_Email	Unique	Email	Email addresses must be unique
FK_Users_Roles	Foreign Key	RoleID	References Roles table
CHK_Users_Email	Check	Email	Email must contain @ and .
DF_Users_IsActive	Default	IsActive	Default to active (1)
DF_Users_CreatedDate	Default	CreatedDate	Default to current date/time
DF_Users_FailedLogin Attempts	Default	FailedLoginAttempts	Default to 0
DF_Users_IsLocked	Default	IsLocked	Default to unlocked (0)

Sample Data

UserID	Email	RoleID	IsActive	IsLocked
1	admin@workops.com	1	1	0
2	manager@workops.com	2	1	0
3	employee1@workops.com	3	1	0

Relationships

Foreign Key	Referenced Table	Referenced Column
FK_Users_Roles	Roles	RoleID
FK_Employees_Users	Employees	UserID
FK_Notifications_Users	Notifications	UserID
FK_ActivityLogs_Users	ActivityLogs	UserID

4.3 Departments Table

Table Description

The Departments table stores organizational department information. Each department has a unique code, name, and optionally a department head. This table forms the organizational hierarchy structure.

Table Schema

Column Name	Data Type	Nullable	Default	Description
DepartmentID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each department
DepartmentName	NVARCHAR(100)	No	None	Full name of the department
DepartmentCode	NVARCHAR(20)	No	None	Short code/abbreviation (unique)
Description	NVARCHAR(500)	Yes	NULL	Detailed description of department functions
DepartmentHeadID	INT	Yes	NULL	Foreign key to Employees table (department head)
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when department was created
IsActive	BIT	Yes	1	Indicates if department is active

Constraints

Constraint Name	Type	Columns	Description
PK_Departments	Primary Key	DepartmentID	Unique identifier for each department
UQ_Departments_DepartmentCode	Unique	DepartmentCode	Department codes must be unique
FK_Departments_DepartmentHead	Foreign Key	DepartmentHeadID	References Employees table
DF_Departments_CreatedDate	Default	CreatedDate	Default to current date/time
DF_Departments_IsActive	Default	IsActive	Default to active (1)

Sample Data

DepartmentID	DepartmentName	DepartmentCode	DepartmentHeadID	IsActive
1	Human Resources	HR	NULL	1
2	Information Technology	IT	2	1
3	Sales & Marketing	SALES	4	1
4	Finance	FIN	NULL	1

5	Operations	OPS	NULL	1
6	Customer Support	SUPPORT	NULL	1

Relationships

Foreign Key	Referenced Table	Referenced Column
FK_Employees_Departments	Employees	DepartmentID
FK_WorkTasks_Departments	WorkTasks	DepartmentID
FK_Departments_DepartmentHead	Employees	DepartmentHeadID

4.4 Employees Table

Table Description

The Employees table stores comprehensive personal and professional information for all employees. This table extends the Users table with additional employee-specific fields.

Table Schema

Column Name	Data Type	Nullable	Default	Description
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EmployeeID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each employee
UserID	INT	No	None	Foreign key to Users table (unique)
FirstName	NVARCHAR(50)	No	None	Employee's first name
LastName	NVARCHAR(50)	No	None	Employee's last name
Phone	NVARCHAR(20)	Yes	NULL	Contact phone number
DepartmentID	INT	No	None	Foreign key to Departments table
ManagerID	INT	Yes	NULL	Self-reference to employee's manager
Position	NVARCHAR(100)	Yes	NULL	Job title/position
JoiningDate	DATE	No	None	Date employee joined the organization
Salary	DECIMAL(18,2)	Yes	NULL	Annual or monthly salary

Address	NVARCHAR (500)	Yes	NULL	Residential address
DateOfBirth	DATE	Yes	NULL	Employee's date of birth
ProfilePicture	NVARCHAR (255)	Yes	NULL	File path to profile picture
EmergencyContact	NVARCHAR (100)	Yes	NULL	Name of emergency contact person
EmergencyPhone	NVARCHAR (20)	Yes	NULL	Phone number of emergency contact
CreatedDate	DATETIME	Yes	GETDATE())	Timestamp when employee record was created
IsActive	BIT	Yes	1	Indicates if employee is currently active

Constraints

Constraint Name	Type	Columns	Description
PK_Employees	Primary Key	EmployeeID	Unique identifier for each employee
UQ_Employees_UserID	Unique	UserID	Each employee has unique UserID

FK_Employees_Users	Foreign Key	UserID	References Users table
FK_Employees_Departments	Foreign Key	DepartmentID	References Departments table
FK_Employees_Managers	Foreign Key	ManagerID	Self-references Employees table
DF_Employees_CreatedDate	Default	CreatedDate	Default to current date/time
DF_Employees_IsActive	Default	IsActive	Default to active (1)

Sample Data

EmployeeID	UserID	FirstName	LastName	DepartmentID	ManagerID	Position	IsActive
1	1	System	Administrator	2	NULL	System Admin	1
2	2	John	Manager	2	NULL	IT Manager	1
3	3	Alice	Johnson	2	2	Software Developer	1

4	4	Bob	Smith	3	NULL	Sales Executive	1
5	5	Carol	Davis	4	NULL	Account ant	1
6	6	David	Wilson	2	2	System Analyst	1

Relationships

Foreign Key	Referenced Table	Referenced Column
FK_Employees_Users	Users	UserID
FK_Employees_Departments	Departments	DepartmentID
FK_Employees_Managers	Employees	ManagerID
FK_WorkTasks_AssignedTo	WorkTasks	AssignedTo
FK_WorkTasks_AssignedBy	WorkTasks	AssignedBy
FK_Attendance_Employees	Attendance	EmployeeID
FK_LeaveRequests_Employees	LeaveRequests	EmployeeID
FK_LeaveBalances_Employees	LeaveBalances	EmployeeID
FK_PerformanceReviews_Employees	PerformanceReviews	EmployeeID

4.5 WorkTasks Table

Table Description

The WorkTasks table stores all task-related information. Tasks can be assigned to employees, have priorities, statuses, due dates, and track completion. The table name uses "WorkTasks" to avoid conflict with the SQL keyword "TASKS".

Table Schema

Column Name	Data Type	Nullable	Default	Description
TaskID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each task
TaskCode	NVARCHAR(50)	No	None	Auto-generated unique task code (TASK-YYYYMM-XXX X)
Title	NVARCHAR(200)	No	None	Task title/summary
Description	NVARCHAR(MAX)	Yes	NULL	Detailed task description

Priority	INT	No	2	Priority level: 1=Low, 2=Medium, 3=High, 4=Critical
Status	INT	No	1	Status level: 1=Pending, 2=In Progress, 3=Submitted, 4=Approved, 5=Completed
AssignedTo	INT	No	None	Foreign key to Employees (who task is assigned to)
AssignedBy	INT	No	None	Foreign key to Employees (who created/assigned task)
DepartmentID	INT	No	None	Foreign key to Departments
StartDate	DATE	Yes	NULL	Date task work begins
DueDate	DATE	No	None	Deadline for task completion

CompletedDate	DATETIME	Yes	NULL	Timestamp when task was marked completed
EstimatedHours	DECIMAL(5, 2)	Yes	NULL	Estimated hours to complete task
ActualHours	DECIMAL(5, 2)	Yes	NULL	Actual hours spent on task
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when task was created
LastUpdatedDate	DATETIME	Yes	GETDATE()	Timestamp when task was last modified

Constraints

Constraint Name	Type	Columns	Description
PK_WorkTasks	Primary Key	TaskID	Unique identifier for each task
UQ_WorkTasks_TaskCode	Unique	TaskCode	Task codes must be unique
FK_WorkTasks_AssignedTo	Foreign Key	AssignedTo	References Employees table

FK_WorkTasks_AssignedBy	Foreign Key	AssignedBy	References Employees table
FK_WorkTasks_Departments	Foreign Key	DepartmentID	References Departments table
CHK_WorkTasks_Priority	Check	Priority	Priority must be between 1 and 4
CHK_WorkTasks_Status	Check	Status	Status must be between 1 and 5
CHK_WorkTasks_DueDate	Check	DueDate, StartDate	DueDate must be >= StartDate
DF_WorkTasks_Priority	Default	Priority	Default to 2 (Medium)
DF_WorkTasks_Status	Default	Status	Default to 1 (Pending)
DF_WorkTasks_CreatedDate	Default	CreatedDate	Default to current date/time
DF_WorkTasks_LastUpdatedDate	Default	LastUpdatedDate	Default to current date/time

Sample Data

TaskID	TaskCode	Title	Priority	Status	AssignedTo	DueDate
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1	TASK-001	Design Database Schema	3	2	3	2024-01-10
2	TASK-002	Implement User Authentication	3	2	3	2024-01-15
3	TASK-003	Sales Report Generation	2	1	4	2024-01-12
4	TASK-004	Financial Audit Preparation	4	1	5	2024-01-20
5	TASK-005	System Performance Optimization	3	3	6	2024-01-08
6	TASK-006	Employee Onboarding Portal	2	5	3	2023-12-31

Relationships

Foreign Key	Referenced Table	Referenced Column
FK_TaskComments_WorkTasks	TaskComments	TaskID
FK_TaskDocuments_WorkTasks	TaskDocuments	TaskID

4.6 TaskComments Table

Table Description

The TaskComments table stores comments and discussions related to specific tasks. This enables collaboration between employees working on tasks.

Table Schema

Column Name	Data Type	Nullable	Default	Description
CommentID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each comment
TaskID	INT	No	None	Foreign key to WorkTasks table
EmployeeID	INT	No	None	Foreign key to Employees (comment author)
CommentText	NVARCHAR(MAX)	No	None	Content of the comment
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when comment was created

IsEdited	BIT	Yes	0	Indicates if comment was edited after creation
EditedDate	DATETIME	Yes	NULL	Timestamp when comment was last edited

Constraints

Constraint Name	Type	Columns	Description
PK_TaskComments	Primary Key	CommentID	Unique identifier for each comment
FK_TaskComments_WorkTasks	Foreign Key	TaskID	References WorkTasks table
FK_TaskComments_Employees	Foreign Key	EmployeeID	References Employees table
DF_TaskComments_CreatedDate	Default	CreatedDate	Default to current date/time
DF_TaskComments_IsEdited	Default	IsEdited	Default to not edited (0)

Relationships

Foreign Key	Referenced Table	Referenced Column
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FK_TaskComments_WorkTasks	WorkTasks	TaskID
FK_TaskComments_Employees	Employees	EmployeeID

4.7 TaskDocuments Table

Table Description

The TaskDocuments table stores file attachments associated with tasks. This allows employees to upload supporting documents, screenshots, or other files related to task work.

Table Schema

Column Name	Data Type	Nullable	Default	Description
DocumentID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each document
TaskID	INT	No	None	Foreign key to WorkTasks table
EmployeeID	INT	No	None	Foreign key to Employees (uploader)
DocumentName	NVARCHAR (255)	No	None	Original name of the uploaded file

FilePath	NVARCHAR (500)	No	None	Server file path where document is stored
FileSize	BIGINT	Yes	NULL	Size of file in bytes
FileType	NVARCHAR (50)	Yes	NULL	MIME type or file extension
UploadDate	DATETIME	Yes	GETDATE()	Timestamp when document was uploaded
Description	NVARCHAR (500)	Yes	NULL	Optional description of the document

Constraints

Constraint Name	Type	Columns	Description
PK_TaskDocuments	Primary Key	DocumentID	Unique identifier for each document
FK_TaskDocuments_WorkTasks	Foreign Key	TaskID	References WorkTasks table
FK_TaskDocuments_Employees	Foreign Key	EmployeeID	References Employees table
DF_TaskDocuments_UploadDate	Default	UploadDate	Default to current date/time

Allowed File Types

File Type	Extensions	Max Size
Documents	.pdf, .doc, .docx	10 MB
Spreadsheets	.xls, .xlsx	10 MB
Images	.jpg, .jpeg, .png	10 MB

4.8 Attendance Table

Table Description

The Attendance table stores daily check-in and check-out records for employees. It tracks working hours, late arrivals, overtime, and attendance status.

Table Schema

Column Name	Data Type	Nullable	Default	Description
AttendanceID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each attendance record
EmployeeID	INT	No	None	Foreign key to Employees table
Attendance Date	DATE	No	None	Date of attendance (cannot be future)

CheckInTime	DATETIME	Yes	NULL	Timestamp when employee checked in
CheckOutTime	DATETIME	Yes	NULL	Timestamp when employee checked out
TotalHours	DECIMAL(5, 2)	Yes	NULL	Calculated total working hours for the day
Status	INT	No	1	Status: 1=Present, 2=Absent, 3=Half Day, 4=On Leave, 5=Holiday
LateMinutes	INT	Yes	0	Minutes late beyond threshold (default 9:30 AM)
OvertimeHours	DECIMAL(5, 2)	Yes	0	Hours worked beyond normal workday (8 hours)
Notes	NVARCHAR (500)	Yes	NULL	Additional notes or remarks
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when record was created

Constraints

Constraint Name	Type	Columns	Description
PK_Attendance	Primary Key	AttendanceID	Unique identifier for each attendance record
UQ_Attendance_EmployeeDate	Unique	EmployeeID, AttendanceDate	One record per employee per day
FK_Attendance_Employees	Foreign Key	EmployeeID	References Employees table
CHK_Attendance_Status	Check	Status	Status must be between 1 and 5
DF_Attendance_Status	Default	Status	Default to 1 (Present)
DF_Attendance_LateMinutes	Default	LateMinutes	Default to 0
DF_Attendance_OvertimeHours	Default	OvertimeHours	Default to 0
DF_Attendance_CreatedDate	Default	CreatedDate	Default to current date/time

Sample Data

AttendanceID	EmployeeID	AttendanceDate	CheckIn Time	CheckOutTime	Status	LateMinutes
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1	3	2024-01-01	09:00:00	17:00:00	1	0
2	4	2024-01-01	09:15:00	17:30:00	1	15
3	5	2024-01-01	09:05:00	16:45:00	1	5
4	3	2024-01-02	09:10:00	17:15:00	1	10
5	4	2024-01-02	09:00:00	17:00:00	1	0

4.9 LeaveTypes Table

Table Description

The LeaveTypes table defines the different categories of leave available to employees. Each leave type has a maximum allocation per year and may require supporting documentation.

Table Schema

Column Name	Data Type	Nullable	Default	Description
LeaveTypeID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each leave type

TypeName	NVARCHAR (50)	No	None	Name of leave type (unique)
Description	NVARCHAR (200)	Yes	NULL	Detailed description of leave type
MaxDaysPerYear	INT	No	None	Maximum days allowed per year
IsPaid	BIT	Yes	1	Indicates if leave is paid
RequiresDocument	BIT	Yes	0	Indicates if supporting document is required
CreatedDate	DATETIME	Yes	GETDATE())	Timestamp when leave type was created

Constraints

Constraint Name	Type	Columns	Description
PK_LeaveTypes	Primary Key	LeaveTypeID	Unique identifier for each leave type
UQ_LeaveTypes_TypeName	Unique	TypeName	Leave type names must be unique
DF_LeaveTypes_IsPaid	Default	IsPaid	Default to paid (1)

DF_LeaveTypes_RequiresDocument	Default	RequiresDocument	Default to not required (0)
DF_LeaveTypes_CreatedDate	Default	CreatedDate	Default to current date/time

Sample Data

LeaveTypeID	TypeName	MaxDaysPerYear	IsPaid	RequiresDocument
1	Casual Leave	12	1	0
2	Sick Leave	10	1	1
3	Paid Leave	15	1	0
4	Maternity Leave	180	1	1
5	Paternity Leave	15	1	1
6	Unpaid Leave	30	0	0

4.10 LeaveRequests Table

Table Description

The LeaveRequests table stores employee leave applications. Each request goes through an approval workflow with status tracking.

Table Schema

Column Name	Data Type	Nullable	Default	Description
LeaveRequestID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each leave request
EmployeeID	INT	No	None	Foreign key to Employees (applicant)
LeaveTypeID	INT	No	None	Foreign key to LeaveTypes
StartDate	DATE	No	None	First day of leave
EndDate	DATE	No	None	Last day of leave
TotalDays	INT	No	None	Calculated total working days (excludes weekends)
Reason	NVARCHAR(500)	Yes	NULL	Reason for leave application
Status	INT	No	1	Status: 1=Pending, 2=Approved,

				3=Rejected, 4=Cancelled
ApprovedBy	INT	Yes	NULL	Foreign key to Employees (approver)
ApprovedDate	DATETIME	Yes	NULL	Timestamp when leave was approved/rejected
RejectionReason	NVARCHAR (500)	Yes	NULL	Reason provided if leave is rejected
AttachmentPath	NVARCHAR (500)	Yes	NULL	Path to supporting document if required
AppliedDate	DATETIME	Yes	GETDATE()	Timestamp when application was submitted

Constraints

Constraint Name	Type	Columns	Description
PK_LeaveRequests	Primary Key	LeaveRequestID	Unique identifier for each leave request
FK_LeaveRequests_Employees	Foreign Key	EmployeeID	References Employees table

FK_LeaveRequests_LeaveTypes	Foreign Key	LeaveTypeID	References LeaveTypes table
FK_LeaveRequests_ApprovedBy	Foreign Key	ApprovedBy	References Employees table
CHK_LeaveRequests_EndDate	Check	EndDate, StartDate	EndDate must be >= StartDate
CHK_LeaveRequests_Status	Check	Status	Status must be between 1 and 4
DF_LeaveRequests_Status	Default	Status	Default to 1 (Pending)
DF_LeaveRequests_AppliedDate	Default	AppliedDate	Default to current date/time

4.11 LeaveBalances Table

Table Description

The LeaveBalances table tracks each employee's leave allocation and usage per leave type per year. The RemainingDays column is a calculated column.

Table Schema

Column Name	Data Type	Nullable	Default	Description
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BalanceID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each balance record
EmployeeID	INT	No	None	Foreign key to Employees
LeaveTypeID	INT	No	None	Foreign key to LeaveTypes
Year	INT	No	None	Calendar year for which balance applies
TotalDays	INT	No	None	Total days allocated for the year
UsedDays	INT	Yes	0	Days already used/approved
Remaining Days	INT	Computed	TotalDays - UsedDays	Automatically calculated remaining days

Constraints

Constraint Name	Type	Columns	Description
PK_LeaveBalances	Primary Key	BalanceID	Unique identifier for each balance record

UQ_LeaveBalance_EmployeeTypeYear	Unique	EmployeeID, LeaveTypeID, Year	One balance record per employee per type per year
FK_LeaveBalances_Employees	Foreign Key	EmployeeID	References Employees table
FK_LeaveBalances_LeaveTypes	Foreign Key	LeaveTypeID	References LeaveTypes table
DF_LeaveBalances_UsedDays	Default	UsedDays	Default to 0

Sample Data

BalanceID	EmployeeID	LeaveTypeID	Year	TotalDays	UsedDays	RemainingDays
1	3	1	2024	12	2	10
2	3	2	2024	10	0	10
3	3	3	2024	15	0	15
4	4	1	2024	12	0	12
5	4	2	2024	10	3	7

4.12 Notifications Table

Table Description

The Notifications table stores in-app notifications for users. Notifications are generated for task assignments, leave approvals, deadline reminders, and other system events.

Table Schema

Column Name	Data Type	Nullable	Default	Description
NotificationID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each notification
UserID	INT	No	None	Foreign key to Users (recipient)
Title	NVARCHAR (200)	No	None	Short notification title
Message	NVARCHAR (500)	No	None	Detailed notification message
Notification Type	INT	No	None	Type: 1=Task, 2=Leave, 3=Attendance, 4=System, 5=Alert
IsRead	BIT	Yes	0	Indicates if user has read the notification

Link	NVARCHAR (500)	Yes	NULL	URL to related page/resource
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when notification was created
ReadDate	DATETIME	Yes	NULL	Timestamp when user read the notification

Constraints

Constraint Name	Type	Columns	Description
PK_Notifications	Primary Key	NotificationID	Unique identifier for each notification
FK_Notifications_Users	Foreign Key	UserID	References Users table
DF_Notifications_IsRead	Default	IsRead	Default to unread (0)
DF_Notifications_CreateDate	Default	CreatedDate	Default to current date/time

4.13 ActivityLogs Table

Table Description

The ActivityLogs table provides an audit trail of all significant user actions in the system. This is critical for security, compliance, and troubleshooting.

Table Schema

Column Name	Data Type	Nullable	Default	Description
LogID	BIGINT	No	IDENTITY(1,1)	Primary key, unique identifier for each log entry
UserID	INT	No	None	Foreign key to Users (actor)
ActivityType	NVARCHAR(100)	No	None	Type of activity (e.g., LOGIN, CREATE, UPDATE, DELETE)
Description	NVARCHAR(MAX)	No	None	Detailed description of the action
IPAddress	NVARCHAR(50)	Yes	NULL	IP address of the user
UserAgent	NVARCHAR(500)	Yes	NULL	Browser/device information
LogDate	DATETIME	Yes	GETDATE()	Timestamp when activity occurred

AffectedRecordID	INT	Yes	NULL	ID of the record affected
AffectedTable	NVARCHAR(100)	Yes	NULL	Name of table containing affected record

Constraints

Constraint Name	Type	Columns	Description
PK_ActivityLogs	Primary Key	LogID	Unique identifier for each log entry
FK_ActivityLogs_Users	Foreign Key	UserID	References Users table
DF_ActivityLogs_LogDate	Default	LogDate	Default to current date/time

Indexes for Performance

Index Name	Columns	Purpose
IX_ActivityLogs_LogDate	LogDate	Date range queries
IX_ActivityLogs_UserId	UserID	User-specific logs
IX_ActivityLogs_ActivityType	ActivityType	Activity type filtering

4.14 Holidays Table

Table Description

The Holidays table stores company holiday information. Holidays are used in attendance calculations and leave management.

Table Schema

Column Name	Data Type	Nullable	Default	Description
HolidayID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each holiday
HolidayName	NVARCHAR(100)	No	None	Name of the holiday
HolidayDate	DATE	No	None	Date of the holiday (unique)
Description	NVARCHAR(500)	Yes	NULL	Additional information about the holiday
IsRecurring	BIT	Yes	1	Indicates if holiday occurs every year
CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when holiday was created

Constraints

Constraint Name	Type	Columns	Description
PK_Holidays	Primary Key	HolidayID	Unique identifier for each holiday
UQ_Holidays_HolidayD	Unique	HolidayDate	One holiday per date
DF_Holidays_IsRecurri	Default	IsRecurring	Default to recurring (1)
DF_Holidays_CreatedD	Default	CreatedDate	Default to current date/time

Sample Data

HolidayID	HolidayName	HolidayDate	IsRecurring
1	New Year's Day	2024-01-01	1
2	Republic Day	2024-01-26	1
3	Holi	2024-03-25	0
4	Independence Day	2024-08-15	1
5	Diwali	2024-11-12	0

4.15 Settings Table

Table Description

The Settings table stores system configuration as key-value pairs. This allows administrators to modify system behavior without code changes.

Table Schema

Column Name	Data Type	Nullable	Default	Description
SettingID	INT	No	IDENTITY(1,1)	Primary key, unique identifier for each setting
SettingKey	NVARCHAR(100)	No	None	Unique key identifier for the setting
SettingValue	NVARCHAR(MAX)	Yes	NULL	Value associated with the setting
SettingType	NVARCHAR(50)	Yes	NULL	Data type (String, Int, Bool, Time)
Category	NVARCHAR(100)	Yes	NULL	Category grouping for settings
Description	NVARCHAR(500)	Yes	NULL	Description of what the setting controls

CreatedDate	DATETIME	Yes	GETDATE()	Timestamp when setting was created
UpdatedDate	DATETIME	Yes	GETDATE()	Timestamp when setting was last modified

Constraints

Constraint Name	Type	Columns	Description
PK_Settings	Primary Key	SettingID	Unique identifier for each setting
UQ_Settings_SettingKey	Unique	SettingKey	Setting keys must be unique
DF_Settings_CreatedDate	Default	CreatedDate	Default to current date/time
DF_Settings_UpdatedDate	Default	UpdatedDate	Default to current date/time

Sample Data

SettingKey	SettingValue	SettingType	Category
CompanyName	WorkOps Inc.	String	General
CompanyEmail	info@workops.com	String	General

WorkStartTime	09:00	Time	Attendance
WorkEndTime	18:00	Time	Attendance
LateThreshold	09:30	Time	Attendance
MaxFileSize	10485760	Int	File
AllowedFileTypes	pdf,doc,docx,xls,xlsx,jpg,jpeg,png	String	File
SmtpServer	smtp.gmail.com	String	Email
LeaveApprovalRequired	true	Bool	Leave

4.16 PerformanceReviews Table

Table Description

The PerformanceReviews table stores employee performance evaluation records. Reviews can be conducted periodically and include ratings, comments, and goals.

Table Schema

Column Name	Data Type	Nullable	Default	Description
ReviewID	INT	No	IDENTITY (1,1)	Primary key, unique identifier for each review

EmployeeID	INT	No	None	Foreign key to Employees (reviewee)
ReviewDate	DATE	No	None	Date when review was conducted
ReviewerID	INT	No	None	Foreign key to Employees (reviewer)
Rating	DECIMAL(3, 1)	No	None	Rating from 1.0 to 5.0
Comments	NVARCHAR (MAX)	Yes	NULL	Overall review comments
Strengths	NVARCHAR (MAX)	Yes	NULL	Employee's strengths identified
AreasForImprovement	NVARCHAR (MAX)	Yes	NULL	Areas where employee can improve
Goals	NVARCHAR (MAX)	Yes	NULL	Goals for next review period
NextReviewDate	DATE	Yes	NULL	Suggested date for next review
CreatedDate	DATETIME	Yes	GETDATE ()	Timestamp when review was created

Constraints

Constraint Name	Type	Columns	Description
PK_PerformanceReviews	Primary Key	ReviewID	Unique identifier for each review
FK_PerformanceReviews_Employees	Foreign Key	EmployeeID	References Employees table
FK_PerformanceReviews_Reviewers	Foreign Key	ReviewerID	References Employees table
CHK_PerformanceReviews_Rating	Check	Rating	Rating must be between 1 and 5
DF_PerformanceReviews_CreatedDate	Default	CreatedDate	Default to current date/time

SECTION 5: STORED PROCEDURES DOCUMENTATION

5.1 sp_AuthenticateUser

Purpose

Validates user login credentials, checks account status, and handles failed login attempts and account locking.

Parameters

Parameter	Data Type	Direction	Description
@Email	NVARCHAR(100)	Input	User's email address
@Password	NVARCHAR(255)	Input	User's password (plain text, will be hashed)

Return Values

Result Value	Description
Success	Authentication successful, returns user details
UserNotFound	No user exists with provided email
AccountLocked	Account is locked due to too many failed attempts
AccountInactive	Account exists but is deactivated
InvalidPassword	Password does not match

Logic Flow

1. Look up user by email
2. Check if user exists
3. Check if account is locked
4. Check if account is active
5. Compare provided password with stored hash
6. On success: update LastLoginDate, reset failed attempts
7. On failure: increment FailedLoginAttempts, lock if exceeds 5
8. Return appropriate result with user information

5.2 sp_MarkAttendance

Purpose

Processes check-in and check-out operations for employees, calculating hours, late minutes, and overtime.

Parameters

Parameter	Data Type	Direction	Description
@EmployeeID	INT	Input	ID of employee marking attendance
@Action	NVARCHAR(10)	Input	CHECKIN or CHECKOUT

Return Values

Column	Description
Result	Success or error message
AttendanceDate	Date of the attendance record

Logic Flow for CHECKIN

1. Get today's date
2. Check if attendance record exists for today
3. If not, create new record with CheckInTime
4. If exists, update CheckInTime
5. Calculate late minutes if check-in after 9:30 AM

Logic Flow for CHECKOUT

1. Get today's date
 2. Find attendance record for today
 3. Update CheckOutTime
 4. Calculate TotalHours if CheckInTime exists
 5. Calculate OvertimeHours if TotalHours > 8
-

5.3 sp_GetAdminDashboardData

Purpose

Retrieves all statistics and data needed for the Admin Dashboard display.

Parameters

None

Return Result Sets

Result Set	Description
Employee Stats	Total, active, inactive employee counts
Department Count	Total active departments
Task Stats	Total, pending, in progress, submitted, completed tasks
Attendance Today	Present, absent, half day, on leave counts
Pending Leaves	Count of leave requests with status Pending

Department Distribution	Department names with employee counts
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5.4 sp_GetManagerDashboardData

Purpose

Retrieves statistics and data for a manager's dashboard, filtered to their team.

Parameters

Parameter	Data Type	Direction	Description
@ManagerID	INT	Input	ID of the manager

Return Result Sets

Result Set	Description
Team Size	Number of employees reporting to manager
Team Task Stats	Task counts for team members
Pending Leaves	Leave requests from team awaiting approval
Employee Performance	Task completion metrics per team member

5.5 sp_GetEmployeeDashboardData

Purpose

Retrieves personal statistics and data for an employee's dashboard.

Parameters

Parameter	Data Type	Direction	Description
@EmployeeID	INT	Input	ID of the employee

Return Result Sets

Result Set	Description
Employee Info	Basic employee information
Task Stats	Personal task counts by status
Today's Attendance	Current day's attendance record
Leave Balances	Current year leave balances

SECTION 6: VIEWS DOCUMENTATION

6.1 vw_EmployeeDetails

Purpose

Provides a comprehensive view of employee information with department and manager names, eliminating the need for multiple joins.

Columns

Column Name	Source	Description
EmployeeID	Employees	Unique employee identifier
FullName	Employees	Concatenated first and last name
LoginEmail	Users	Employee's login email
Phone	Employees	Contact phone number
DepartmentName	Departments	Name of employee's department
Position	Employees	Job title
JoiningDate	Employees	Date joined organization
IsActive	Employees	Active status
ManagerName	Employees (self)	Name of employee's manager
RoleName	Roles	User role name

Use Cases

- Employee list display
- Employee details page
- Reports requiring employee information

6.2 vw_TaskDetails

Purpose

Provides a complete view of task information with assignee names, priority text, status text, and deadline calculations.

Columns

Column Name	Source	Description
TaskID	WorkTasks	Unique task identifier
TaskCode	WorkTasks	Auto-generated task code
Title	WorkTasks	Task title
Description	WorkTasks	Detailed description
Priority	WorkTasks	Numeric priority
PriorityText	Calculated	Low, Medium, High, Critical
Status	WorkTasks	Numeric status
StatusText	Calculated	Pending, In Progress, Submitted, Approved, Completed
AssignedTo	Employees	Name of assigned employee
AssignedBy	Employees	Name of assigning employee
DepartmentName	Departments	Department name

StartDate	WorkTasks	Start date
DueDate	WorkTasks	Due date
CompletedDate	WorkTasks	Completion timestamp
DaysRemaining	Calculated	Days until due date
DeadlineStatus	Calculated	Overdue, Due Soon, On Track

Use Cases

- Task list display
- Task details page
- Manager task overview
- Deadline monitoring

6.3 vw_MonthlyAttendance

Purpose

Provides monthly attendance summary per employee with aggregated metrics.

Columns

Column Name	Source	Description
EmployeeID	Employees	Unique employee identifier
EmployeeName	Employees	Full name

DepartmentName	Departments	Department name
Year	Attendance	Calendar year
Month	Attendance	Month number (1-12)
PresentDays	Attendance	Count of Present days
AbsentDays	Attendance	Count of Absent days
HalfDays	Attendance	Count of Half Day
LeaveDays	Attendance	Count of On Leave
TotalHours	Attendance	Sum of total working hours
OvertimeHours	Attendance	Sum of overtime hours

Use Cases

- Monthly attendance reports
- Department attendance analytics
- Employee attendance history

6.4 vw_LeaveBalanceSummary

Purpose

Provides leave balance summary with usage percentage calculation.

Columns

Column Name	Source	Description
EmployeeID	Employees	Unique employee identifier
EmployeeName	Employees	Full name
DepartmentName	Departments	Department name
LeaveType	LeaveTypes	Type of leave
Year	LeaveBalances	Calendar year
TotalDays	LeaveBalances	Total days allocated
UsedDays	LeaveBalances	Days used
RemainingDays	LeaveBalances	Days remaining
UsagePercentage	Calculated	Percentage of days used

Use Cases

- Leave balance reports
- Low balance alerts
- Department leave analytics

SECTION 7: TRIGGERS DOCUMENTATION

7.1 trg_Employees_CreateLeaveBalance

Trigger Type

AFTER INSERT on Employees table

Purpose

Automatically creates leave balance records for a new employee for all existing leave types for the current year.

Logic

1. Get the newly inserted EmployeeID
2. Get current year
3. For each leave type in LeaveTypes table
4. Insert a leave balance record with TotalDays = MaxDaysPerYear, UsedDays = 0

Business Impact

Ensures every employee has leave balances automatically without manual intervention.

7.2 trg_LeaveRequests_UpdateLeaveBalance

Trigger Type

AFTER UPDATE on LeaveRequests table

Purpose

Updates employee leave balance when a leave request is approved.

Logic

1. Detect if Status column was updated
2. Check if status changed from Pending (1) to Approved (2)
3. If yes, add TotalDays to UsedDays in LeaveBalances table

Business Impact

Ensures leave balances are always accurate and reflects approved leaves.

7.3 trg_WorkTasks_UpdateCompletedDate

Trigger Type

AFTER UPDATE on WorkTasks table

Purpose

Automatically sets CompletedDate when task status changes to Completed, and clears it if status changes away from Completed.

Logic

1. Detect if Status column was updated
2. If new status is Completed (5), set CompletedDate to GETDATE()
3. If old status was Completed and new status is not, set CompletedDate to NULL
4. Always update LastUpdatedDate to GETDATE()

Business Impact

Ensures accurate tracking of task completion dates without manual entry.

7.4 trg_WorkTasks_ActivityLog

Trigger Type

AFTER INSERT, UPDATE, DELETE on WorkTasks table

Purpose

Logs all task-related activities for audit purposes.

Logic

1. Determine operation type (INSERT, UPDATE, DELETE)
2. Get affected TaskID
3. Create appropriate description
4. Insert record into ActivityLogs table

Business Impact

Provides complete audit trail for all task changes, essential for compliance.

SECTION 8: FUNCTIONS DOCUMENTATION

8.1 fn_CalculateExperience

Purpose

Calculates an employee's total work experience in years, months, and days.

Parameters

Parameter	Data Type	Description
@EmployeeID	INT	ID of the employee

Return Value

NVARCHAR(100) in format "X Years, Y Months, Z Days"

Logic

1. Get JoiningDate from Employees table
 2. Calculate difference from current date
 3. Format as years, months, days
-

8.2 fn_GetPriorityColor

Purpose

Returns Bootstrap color class for task priority levels.

Parameters

Parameter	Data Type	Description
@Priority	INT	Priority value (1-4)

Return Value

NVARCHAR(20) - Bootstrap color class

Mapping

Priority	Return Value
1 (Low)	success
2 (Medium)	info
3 (High)	warning

4 (Critical)

danger

8.3 fn_GetStatusText

Purpose

Returns human-readable status text for task status codes.

Parameters

Parameter	Data Type	Description
@Status	INT	Status value (1-5)

Return Value

NVARCHAR(20) - Status description

Mapping

Status	Return Value
1	Pending
2	In Progress
3	Submitted
4	Approved

SECTION 9: INDEXES DOCUMENTATION

9.1 Performance Indexes

Table	Index Name	Columns	Purpose
Users	IX_Users_Email	Email	Fast login lookups
Users	IX_Users_RoleID	RoleID	Role-based queries
Employees	IX_Employees_UserID	UserID	User-employee joins
Employees	IX_Employees_Departm entID	DepartmentID	Department filtering
Employees	IX_Employees_ManagerI D	ManagerID	Manager team queries
Employees	IX_Employees_IsActive	IsActive	Active employee filtering
WorkTasks	IX_WorkTasks_Assigned To	AssignedTo	Employee task lists
WorkTasks	IX_WorkTasks_Status	Status	Status filtering
WorkTasks	IX_WorkTasks_DueDate	DueDate	Deadline queries

Attendance	IX_Attendance_EmployeeID	EmployeeID	Employee attendance
Attendance	IX_Attendance_AttendanceDate	AttendanceDate	Date range queries
LeaveRequests	IX_LeaveRequests_EmployeeID	EmployeeID	Employee leave history
LeaveRequests	IX_LeaveRequests_Status	Status	Pending approvals
Notifications	IX_Notifications_UserID	UserID	User notifications
Notifications	IX_Notifications_IsRead	IsRead	Unread count
ActivityLogs	IX_ActivityLogs_UserID	UserID	User activity
ActivityLogs	IX_ActivityLogs_LogDate	LogDate	Date range audit
ActivityLogs	IX_ActivityLogs_ActivityType	ActivityType	Activity filtering

SECTION 10: RELATIONSHIPS AND CONSTRAINTS

10.1 Entity Relationship Summary

Parent Table	Child Table	Relationship Type
Roles	Users	One to Many

Users	Employees	One to One
Users	Notifications	One to Many
Users	ActivityLogs	One to Many
Departments	Employees	One to Many
Departments	WorkTasks	One to Many
Employees	WorkTasks (AssignedTo)	One to Many
Employees	WorkTasks (AssignedBy)	One to Many
Employees	Attendance	One to Many
Employees	LeaveRequests	One to Many
Employees	LeaveBalances	One to Many
Employees	PerformanceReviews (Employee)	One to Many
Employees	PerformanceReviews (Reviewer)	One to Many
Employees	TaskComments	One to Many
Employees	TaskDocuments	One to Many
WorkTasks	TaskComments	One to Many
WorkTasks	TaskDocuments	One to Many
LeaveTypes	LeaveRequests	One to Many

LeaveTypes	LeaveBalances	One to Many
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10.2 Cascading Rules

Constraint	Action
FK_WorkTasks_AssignedTo	NO ACTION (prevent orphaned tasks)
FK_WorkTasks_AssignedBy	NO ACTION
FK_LeaveRequests_Employees	CASCADE DELETE (if employee deleted)
FK_Attendance_Employees	CASCADE DELETE
FK_Notifications_Users	CASCADE DELETE

SECTION 11: SEED DATA

11.1 Default Roles

RoleID	RoleName	Description
1	Admin	System Administrator with full access
2	Manager	Department Manager with team management access

3	Employee	Regular employee with limited access
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11.2 Default Leave Types

LeaveTypeID	TypeName	MaxDaysPerYear	IsPaid	RequiresDocument
1	Casual Leave	12	1	0
2	Sick Leave	10	1	1
3	Paid Leave	15	1	0
4	Maternity Leave	180	1	1
5	Paternity Leave	15	1	1
6	Unpaid Leave	30	0	0

11.3 Default Departments

DepartmentID	DepartmentName	DepartmentCode
1	Human Resources	HR

2	Information Technology	IT
3	Sales & Marketing	SALES
4	Finance	FIN
5	Operations	OPS
6	Customer Support	SUPPORT

11.4 Default Users and Employees

Email	Password	Role	First Name	Last Name	Department
admin@workops.com	Admin@123	Admin	System	Administrator	IT
manager@workops.com	Manager@123	Manager	John	Manager	IT
employee1@workops.com	Employee@123	Employee	Alice	Johnson	IT
employee2@workops.com	Employee@123	Employee	Bob	Smith	Sales
employee3@workops.com	Employee@123	Employee	Carol	Davis	Finance

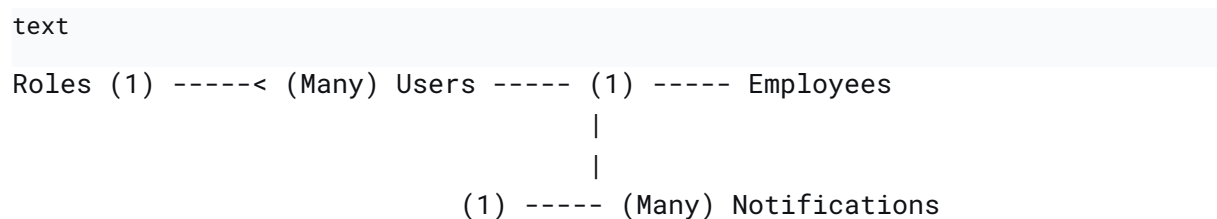
employee4@work ops.com	Employee e@123	Employee e	David Wilson	IT
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11.5 Default System Settings

SettingKey	SettingValue	Category
CompanyName	WorkOps Inc.	General
CompanyEmail	info@workops.com	General
WorkStartTime	09:00	Attendance
WorkEndTime	18:00	Attendance
LateThreshold	09:30	Attendance
MaxFileSize	10485760	File
LeaveApprovalRequired	true	Leave

SECTION 12: ENTITY RELATIONSHIP DIAGRAM DESCRIPTION

12.1 Core Structure (Authentication)



|

(1) ----- (Many) ActivityLogs

Description: The Roles table defines user types. Each User belongs to one Role. Each User has exactly one Employee record. Users can have multiple Notifications and multiple ActivityLog entries.

12.2 Organizational Structure

text

Departments (1) -----< (Many) Employees

|
| (Self-reference for Department Head)
|

----- (1) ----- Employees

Description: Each Department can have multiple Employees. Each Employee belongs to one Department. A Department can have one Department Head who is an Employee.

12.3 Task Management

text

Employees (1) -----< (Many) WorkTasks (AssignedTo)

Employees (1) -----< (Many) WorkTasks (AssignedBy)

Departments (1) -----< (Many) WorkTasks

WorkTasks (1) -----< (Many) TaskComments

WorkTasks (1) -----< (Many) TaskDocuments

Description: Tasks are assigned to and created by Employees. Each Task belongs to one Department. Tasks can have multiple Comments and multiple Documents.

12.4 Attendance Tracking

text

Employees (1) -----< (Many) Attendance

Description: Each Employee has multiple Attendance records (one per working day). The combination of EmployeeID and AttendanceDate is unique.

12.5 Leave Management

text		
Employees	(1) -----< (Many)	LeaveRequests
LeaveTypes	(1) -----< (Many)	LeaveRequests
Employees	(1) -----< (Many)	LeaveBalances
LeaveTypes	(1) -----< (Many)	LeaveBalances
Employees	(1) -----< (Many)	LeaveRequests (ApprovedBy)

Description: Each Employee can have multiple LeaveRequests and multiple LeaveBalances (one per leave type per year). Each LeaveRequest has one LeaveType. LeaveRequests can be approved by one Employee (manager).

12.6 Performance Reviews

text		
Employees	(1) -----< (Many)	PerformanceReviews (Reviewee)
Employees	(1) -----< (Many)	PerformanceReviews (Reviewer)

Description: Each Employee can have multiple Performance Reviews as the reviewee and can also conduct multiple reviews as the reviewer.

SECTION 13: DATABASE SCRIPTS SUMMARY

13.1 Scripts Included

Script Type	Count	Purpose
CREATE TABLE	14	Table creation statements

ALTER TABLE (FK)	20+	Foreign key constraints
CREATE INDEX	25+	Performance indexes
CREATE PROCEDURE	5	Stored procedures
CREATE VIEW	4	Database views
CREATE TRIGGER	4	Automated triggers
CREATE FUNCTION	3	Scalar functions
INSERT (Seed)	6	Initial data population

13.2 Script Execution Order

1. CREATE DATABASE
2. CREATE TABLES (in dependency order)
3. ALTER TABLE ADD FOREIGN KEYS
4. CREATE INDEXES
5. CREATE FUNCTIONS
6. CREATE PROCEDURES
7. CREATE VIEWS
8. CREATE TRIGGERS
9. INSERT SEED DATA

DOCUMENT INFORMATION

Attribute	Detail
Document Version	1.0

Last Updated	April 2026
Database Name	WorkOpsDB
Total Tables	14
Total Columns	100+
Total Relationships	20+

WorkOps – Smart Company Management System
Complete Database Documentation